HOM(F, G) IS THE EQUALIZER OVER AN FPPF COVERING OF X

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Allowed standard prerequisites (editorial).

- Situation situation-hom (canonical tag 08JT). cited at statement line 519. Standing situation; fppf coverings of X and of the pairwise intersections.
- Lemma spaces-topologies-lemma-fppf (canonical tag 03Y9). cited at proof line 543. Base change of an fppf covering is an fppf covering.
- Proposition spaces-descent-proposition-fpqc-descent-quasi-coherent (canonical tag 04W8). cited at proof line 547 (twice). Two applications: agreement on intersections and the unique descent u.

Source reference. Original definition 08JT: sources/stacks-chow/tags.txt, source lines 10861–10861. Cited at statement line: the lemma is phrased in this situation; load-bearing.

Source reference. Original lemma 03Y9: sources/stacks-chow/tags.txt, source lines 5176–5176. Cited at proof line 543: base-changed coverings.

Source reference. Original proposition 04W8: sources/stacks-chow/tags.txt, source lines 6351–6351. Cited at proof line: the descent of the glued morphism; load-bearing.

Lemma 0.2. *In Situation 08JT let $\{X_i \to X\}_{i\in I}$ be an fppf covering and for each $i, j \in I$ let $\{X_{ijk} \to X_i \times_X X_j\}$ be an fppf covering. Denote $\mathcal{F}_i$, resp. $\mathcal{F}_{ijk}$ the pullback of $\mathcal{F}$ to X_i, resp. X_{ijk}. Similarly define $\mathcal{G}_i$ and $\mathcal{G}_{ijk}$. For every scheme T over B the diagram*

$$Hom(\mathcal{F},\mathcal{G})(T) \longrightarrow \prod_i Hom(\mathcal{F}_i,\mathcal{G}_i)(T) \underset{pr_1^*}{\overset{pr_0^*}{\rightrightarrows}} \prod_{i,j,k} Hom(\mathcal{F}_{ijk},\mathcal{G}_{ijk})(T)$$

presents the first arrow as the equalizer of the other two.

Proof. Let $u_i : \mathcal{F}_{i,T} \to \mathcal{G}_{i,T}$ be an element in the equalizer of pr_0^* and pr_1^*. Since the base change of an fppf covering is an fppf covering (Topologies on Spaces, Lemma 03Y9) we see that $\{X_{i,T} \to X_T\}_{i\in I}$ and $\{X_{ijk,T} \to X_{i,T} \times_{X_T} X_{j,T}\}$ are fppf coverings. Applying Descent on Spaces, Proposition 04W8 we first conclude

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that u_i and u_j restrict to the same morphism over $X_{i,T} \times_{X_T} X_{j,T}$, whereupon a second application shows that there is a unique morphism $u : \mathcal{F}_T \to \mathcal{G}_T$ restricting to u_i for each i. This finishes the proof. $\square$